UPSTREAM

Seismology for sewage: find the leak from the first dirty glass

WeMakeDevs Environmental Hacks
Track: Heat & Water · SDG 6
Built on AWS · October 2026

Upstream determines the section of the pipeline through which contaminated effluent reaches a municipal drinking water supply based on the analysis of when the initial tainted cup is served in each house. This requires each house to have three things: a printed card, a chlorine test strip, and a phone. The system runs on AWS without servers in the backend using S3, Lambda, DynamoDB, Step Functions, Bedrock, SNS and Amplify.

1 The problem: sewage in the pipes, and no can tell where

Sewage invaded Bhagirath Pura's tap water in Indore during the end of 2025. This is after all, Indore, the cleanest city in India according to ratings. The residents had complained about the bad taste for several weeks. As at 12th February 2026, the health department had received 35 reports of death out of which 22 were believed to be linked with contaminated water while an investigating committee in October 2026 castigated continued neglect for years.

It works like this everywhere. There are approximately 1 billion individuals worldwide who are experiencing on-and-off tap water at best. The way it happens are as follows: there is a break in between the supply in the pipes which makes them empty out. When the pressure falls, the sewage is sucked into the system through any available entry points, and as the clean water that flow returns, it is used to flush out all the worst impurities first but it arrives at a different moment in every house. The people of Indore get their supply(s) once in two days.

- **Reason why Current Failure of Detection Systems:**
- Even if there is a single leak, five lanes appearance with five problems seeming like one ticket handling approach for complaints fail to work.
- Nobody bothers to monitor if the tap water gets contaminated.
- The pipeline diagrams are often missing many details or very old.
- The laboratory report comes too late after the victims have experienced illness.
- In-pipe IoT sensors requires installation and maintenance at every node

Hackathon fit. Upstream sits in the Heat & Water track and makes water safer. It is targeting leaks, monsoon contamination as well as intermittent supply, with no hardware in the pipes, all while running on AWS as the hackathon requires.

2 The idea: seismology for sewage

Seismologists are used to locate an earthquake from the time where its waves are used reach different stations. Upstream does the same with water. The dirty slug that is around a crack is then pushed downstream when supply resumes, so it reaches nearby homes first and distant homes and later on, more diluted at every junction. Homes upstream of the crack never see it.

This means that the mere complaints of the people cannot be taken as a signal. The signal only becomes real when the tap is dirty with water. On the other hand, there are times when it is possible for very few timestamped readings to serve the purpose that many individual complaints cannot achieve.

Earthquake	Upstream
Epicentre	The section of pipe at which the effluent enters.
Seismograph stations	Individuals capturing pictures of the initial glass at home.
Wave arrival time	The moment the contaminated viscous fluid arrives at the outle
Wave strength	The level of cloudiness and amount of chlorine reduced.
Velocity model	The knowledge acquired on the travelling duration within the piping system

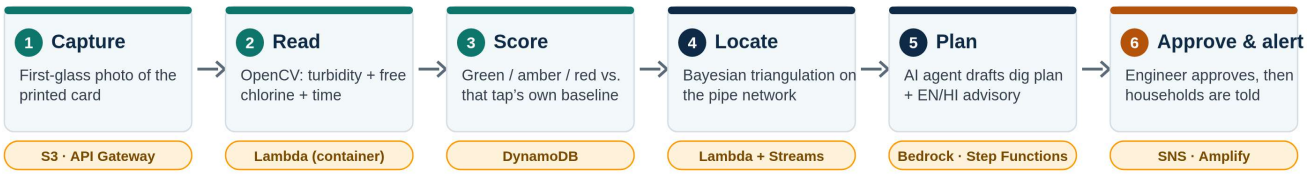


Fig 1. End-to-end flow. The orange tag under each step names the AWS service that runs it.

For the water department	For downstream households	For the planner and the ward
It is possible to export the epicentre data (dig spot, 90% search region, and evidence behind it such as photos, timings, and chlorine levels) in form of CSV or JSON files.	It is also possible to include a personal safe in case the tap water needs boiling, for example, "fill drinking water after 7:14 am" or otherwise, boil water advisory written in two languages English and Hindi. Those living at the upper stream are calm and collected.	The following are the matters to be considered: the homes that will be tested in the subsequent supply, the lane that will be added to recruitment next, and a risk map with respect to water-sewer crossing especially for pre-monsoon

3 How it works: from one photo to a dig spot

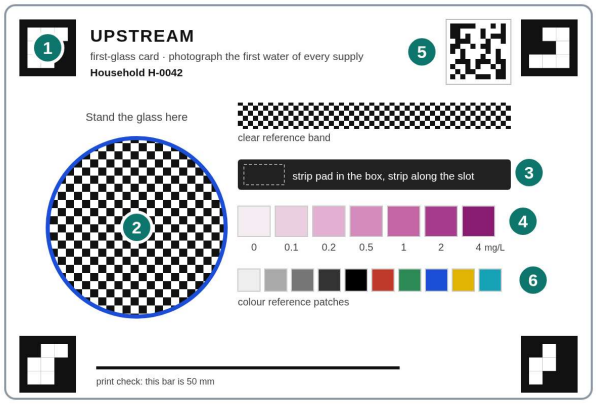


Fig 2. The Upstream card (A5, schematic). Numbers match the list on the right.

1. **Four ArUco corner markers that** undoes the camera angle and distance.
2. **Checkerboard under the glass** blurs just as the water gets cloudier with time.
3. **Strip slot** along with a box for the chlorine pad.
4. **Printed chlorine scale** (0 to 4 mg/L) beside the strip, so both of them are seen in the same light.
5. **QR code** carries only the household ID, no name or phone for privacy.
6. **Colour patches helps in** correction for the lighting.

The first glass protocol. The initial water that comes should be poured in a transparent glass which will then be placed on the circle after taking up a strip laid in the slot but before that one photo will be taken (QR web page or WhatsApp). The photo can have different options such as tap fine, smells bad, looks dirty or someone is ill, however, only "Fine" answers are of importance as they help in determining whether there is any problem with the pipes.

Ten strips are intended for use in regular households, fourteen strips for lanes where people have volunteered and twenty-one strips for those households which act as sentinels and have been chosen by the planner. \ Chlorine Reasoning: It is unbelievable how unsafe tap water appears to be for human consumption when dealing with pathogens in drinking water. It seems that even with all the efforts made by the Effluent Treatment Plants to chlorinate the water and make sure that everything is in place, people cannot still trust it because they think that it may contain some pathogens. The Indian recommends that there should be at least 0. 2 mg/l free residual chlorine level at the consumer end according to IS 10500.

Reading engine (OpenCV)

Reading	How it is measured	How it is used
Time	Photo EXIF or message timestamp, compared with the supply start time	Arrival time of the water, or the dirty slug, at the tap
Turbidity	Holography coming from the markers, colour correction from the patches, then at index = 1 − contrast under glass ÷ contrast of the clear reference	Spikes against that tap's own baseline; not a certificate of the 1 NTU standard
Free chlorine	Strip pad is placed on printed scale in the CIELAB colour space	Below 0.2 mg/L, or a sharp drop from that tap's normal
Quality gate	Blur, glare, overexposure, card-not-found checks; SHA-256 and perceptual hash	Retake requested; re-used photos are flagged and is excluded

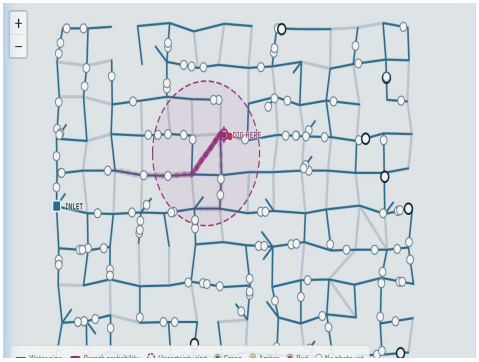
Published work supports both measurements: phone-only turbidity banding reached 91 to 99% accuracy on lab samples, and colour-referenced chlorine strips are about 88% in real lighting.

Signal levels, set against each tap's own baseline

Green	Chlorine ≥ 0.2 mg/L and turbidity normal for that tap	Logged as baseline; timing feeds the network model
Amber	Chlorine < 0.2 mg/L, or turbidity above that tap's normal	Nearby homes with strips are asked to test at the next supply
Red	Near-zero chlorine plus a turbidity spike, or a cluster of smell, colour or illness reports	Triangulation runs, the utility is alerted, downstream homes get safe-after times

Bayesian locator

- **Flow tree. The waterfront arrives at every point on its quickest path once the flow starts. Such paths form a tree even in pipes with loops. Sewage is pushed forward by the front and as a result, it gets into homes located after the breach in a diluted form.**
- **Velocity model. The time it takes for light to get to a house on a normal day is given by photo times. These are converted into traveling times along pipes using a regularized least squares fitting. In the absence of a utility plan, a pipe tree is reconstructed over the OpenStreetMap street grid according to the order of arrival (it recovers about 70% of real pipe length).**
- **Scoring. Every pipe piece of 20 m or less is a candidate. The physical model is combined with a general outlier component in each report, so even if there are pranks or bad photos, they will hardly affect the outcome. The strength of the slug is marginalized since not all cracks need to leak every time there is a supply.**



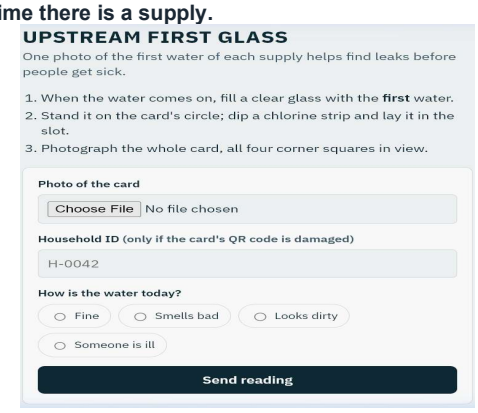


Fig 3 (left). Live intake on the demo ward: a red reading triggers triangulation, with the dig spot and uncertainty ring. Fig 4 (right). The household page opened from the card's QR code. Screens use the offline synthetic ward; the Indore grid loads the same way.

4 Built on AWS: serverless, event-driven, pay-per-use

Every alternative day, supply is brought for nearly two hours. This means that there are high load periods followed by some low or no load ones. This is perfect for serverless because every stage is an AWS service which costs next to nothing when not in use. Regardless of whether it is run on a laptop or within Lambdas that originate from a single container picture on Amazon ECR, the same Python package works flawlessly; and also, because the Indore ward pickle has been inserted into the image, the cold start should be very short. It is all captured in a single AWS SAM template (aws/template. yaml) for the complete stack.

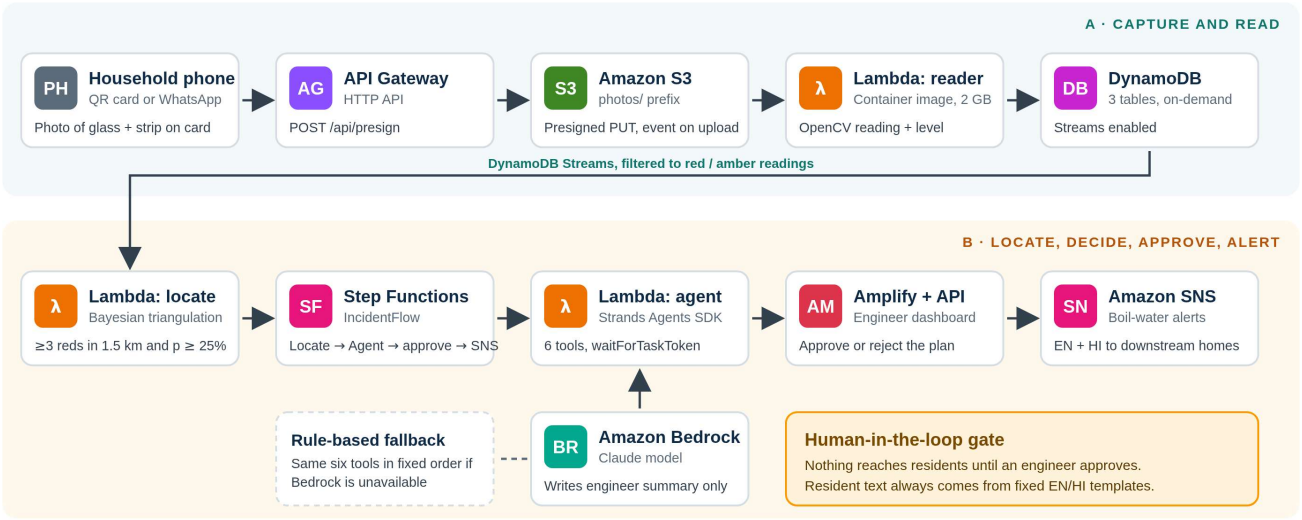


Fig 5. Production architecture. Orange-tagged services are AWS; the dashed box is the fallback path.

AWS service	Role in Upstream
Amazon S3	In the photos directory, photos of first-glass are saved. The households upload through PUT which is pre-assigned after which every upload triggers a reading.
Amazon API Gateway	There is an HTTP API available for presign, results, live state, incidents, as well as engineer Approve / Reject.
AWS Lambda	OpenCV) reader located in an agent with incident status and REST handlers in a single 2GB container image.
Amazon DynamoDB	These tables contain readings, meta and incidents data which are stored based on an on demand storage capacity plan. It has a stream where if anything with red or amber reading is there then it triggers locate.
AWS Step Functions	IncidentFlow: Locate and then Agent takes one hour maximum using a given task token. Approve publishes alerts while reject closes them; timeout expires otherwise.
Amazon Bedrock	Writing of the engineer's incident summary is done using Claude that is driven through Strands Agents SDK.
Amazon SNS	English and Hindi boil-water advisories are published for downstream households after being approved.
AWS Amplify	Using config. js, it directs at API Gateway and hosts the Leaflet utility dashboard.
SAM, ECR, IAM	Infrastructure as code (cfn-lint-compliant), container registry, least-privilege roles according to function.

5 The incident agent and the human approval gate

A red reading helps alert that household at once. A ward-wide **incident** opens only when at least **3 usable red readings** is comeing from homes within **1.5 km** and the locator puts breach probability at **25% or more**. At most there is one incident opens per supply cycle. The agent then is used to prepare the plan with six tools, each a thin wrapper over tested code:

get_incident_readings	Live readings and levels
run_triangulation	Dig spot, 90% region, breach probability
downstream_households	Who is affected and safe-after times
request_strip_tests	Most informative homes for the next test
draft_advisory(lang)	Fixed English and Hindi templates
submit_for_approval	Stores plan and task token

Guardrails

- **Every message must be authorized by an engineer before sending. SNS is published only after the engineer has given his approval which is denoted by the return of Step Functions task token in AWS.**
- **Safety messages cannot be modified by the model. Bedrock only writes the engineer summary part; but the resident text is always fetched from fixed templates.**
- **A Plan for Every Contingency. In case Bedrock is down, there is a rule-based agent that uses the same six instruments in a predetermined sequence to help in creating identical plans.**
- **Privacy and abuse controls. Some examples are pre signed uploads, an engineer's key on Approve / Reject, WhatsApp signature checks, no phone numbers are stored, lane-level location, and photos that are reused are excluded.**

Other interfaces. The same code serves a FastAPI app along a SQLite store for offline demos, a WhatsApp Cloud API webhook, a printable card maker, and a command line (`upstream serve | card | read | simulate | evaluate | export-site | fetch-streets`).

6 Results: 60 simulated outbreaks on Indore's real street grid

97%
Outbreaks detected
29 of 30 detectable

0 of 60
False alarms
alert with no dirty reading downstream

39 m
Median dig-spot error
with the utility's pipe map

93%
Crack inside 90% region
median region 107 m of pipe

The street grid of Bhagirathpura which comprises 540 junctions and stretches over 25. 9 kilometres as obtained from OpenStreetMap is used in simulation. There are also simulated pipes, 140 homes, risk attributes as well as cracks while on the other hand; the physics of the simulator's slug is purposely different from what the locator assumes.

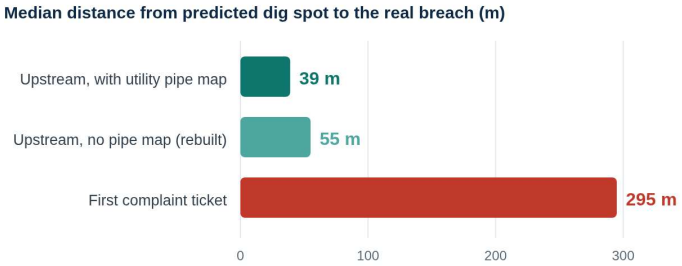


Fig 6. The dig spot lands about 7.5 times closer than the first complaint ticket.

- It takes about 7 photos for the median to reach 50% breach confidence.
- When 10% false red reports are added, the dig spot shifts slightly from 35 m to 43 m.
- In the absence of a pipeline map, there is a median error of 55 m that is covered 72% of the time.
- The flow-tree ensemble increases coverage on the synthetic ward from 75% to 96%.

Measure	Synthetic ward	Indore streets
Detected / detectable	28 / 35	29 / 30
False alarms	1 of 60	0 of 60
Dig-spot error, median	16 m	39 m
Dig-spot error, p90	84 m	114 m
Crack in 90% region	96%	93%
No pipe map, median	21 m	55 m
First complaint ticket	157 m	295 m

7 Cost, feasibility and impact

Item	Cost (estimate)
Laminated A5 card	Rs 2 to 5, once
Chlorine strips, 10-pack	Rs 80 to 120
Phone	Existing; no app, web page via QR
AWS per ward	About US$15 per month, pay-per-use
Per household per year	Under Rs 150

- **No physical installations: there are no sensors, trenching or maintenance contracts.**
- **It does not require a pipeline map as it reconstructs the topology using arrival times.**
- **Double language with human verification: gives advisories in English and Hindi which are first approved by an engineer.**
- **Scale: The fact that intermittent supply affects approximately one billion individuals implies that similar phenomena can be observed across different locations starting from Indore to other cities. Facilitates achievement of the 6th Sustainable Development Goal – universal access to safe drinking water.**

Pilot design: one ward, about 50 volunteer homes, a few 21-strip sentinel homes, one month of baseline before alerts go live. **Measure:** lead time, localisation error, false-alarm rate, strips per confirmed incident, and participation per supply.

8 Status, limitations and roadmap

Piece	Status
Card, reader, levels, travel times, map rebuild, Bayesian locator, advisories, planner	Built and tested
Real Indore street grid (OSM) as default ward	Built
Incident rule, 6-tool agent, EN/HI alerts, Approve / Reject	Built and tested locally
AWS stack (SAM): S3, Lambda, DynamoDB, Step Functions, SNS, Amplify	Written; handlers tested with moto; live deploy next
Real-photo calibration, CloudWatch dashboard	Planned

Stage	Milestone
Field test	Calibrate card and strips on real tap water in varied lighting; fit turbidity to NTU
Ward pilot	50 homes, one month baseline, live alerts with a partner water department
City rollout	Utility GIS integration, risk map, health-department link, Jal Jeevan Mission monitoring

The code and tests comprise a Python package that utilizes FastAPI, OpenCV, NumPy / SciPy, NetworkX and Leaflet. This package includes a pytest suite which is responsible for coverage of the locator, reader, API, agent, DynamoDB store as well as Lambda handlers. On each push, GitHub Action executes this package.

Limitations. The accuracy figures are all based on simulation since the reader has been presented with only synthetic images; however, live images are rated in comparison with simulated archives, whereas stored photographs require a retention policy.

https://github.com/qwertyuiopas17/OverFlow

Upstream: because nobody should die from their first glass of water.

Sources: Kumpel and Nelson 2014; Jantarakasem et al. 2024; González-Gómez et al. 2025; IS 10500:2012; CPHEEO manual; PTI showing the 2025 Indore contamination.